Appendix A
Derivation of Equation (1)

For each DT $n \in \mathcal{N}$ and task type $k \in \mathcal{K}$, the personalized ability improvement in time slot $t \in \mathcal{T}$ can be interpreted as fine-tuning the DT model through gradient descent to minimize a task-dependent loss $\mathcal{F}_{n,k}(\cdot)$ in a continual adaptation process. Let $\boldsymbol{\theta}_n$ denote the current parameter of DT n, and let $\boldsymbol{\theta}^\star_{n,k}$ denote a parameter point specialized for task type k.

It is first shown that the loss $\mathcal{F}_{n,k}(\boldsymbol{\theta}_n)$ exhibits linear/geometric convergence under gradient descent. To this end, the following two standard conditions are imposed on $\mathcal{F}_{n,k}(\cdot)$:

A1. $\mathcal{F}_{n,k}(\cdot)$ is L-smooth. That is, there exists a Lipschitz constant $L > 0$ such that, for any $\boldsymbol{x}$ and $\boldsymbol{y}$,

$$\|\nabla \mathcal{F}_{n,k}(\boldsymbol{x}) - \nabla \mathcal{F}_{n,k}(\boldsymbol{y})\| \le L \|\boldsymbol{x} - \boldsymbol{y}\| . \tag{A.1}$$

A2. $\mathcal{F}_{n,k}(\cdot)$ satisfies the Polyak–Lojasiewicz condition, i.e., for some constant $\mu > 0$ and any $\boldsymbol{x}$,

$$\|\nabla \mathcal{F}_{n,k}(\boldsymbol{x})\|^2 \ge 2\mu \left(\mathcal{F}_{n,k}(\boldsymbol{x}) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})\right) . \tag{A.2}$$

Then, following descent lemma follows from Condition A1.

***Lemma** 1 (Descent Lemma):* For any step size $\delta > 0$ and parameter $\boldsymbol{x}$, the following inequality holds:

$$\begin{aligned} &\mathcal{F}_{n,k}(\boldsymbol{x} - \delta \nabla \mathcal{F}_{n,k}(\boldsymbol{x})) \\ &\le \mathcal{F}_{n,k}(\boldsymbol{x}) - \delta \left(1 - \frac{L\delta}{2}\right) \|\nabla \mathcal{F}_{n,k}(\boldsymbol{x})\|^2 . \end{aligned} \tag{A.3}$$

Next, after one gradient step, the model parameter is given by

$$\boldsymbol{\theta}^1_n = \boldsymbol{\theta}_n - \delta \nabla \mathcal{F}_{n,k}(\boldsymbol{\theta}_n). \tag{A.4}$$

Applying Lemma 1 yields

$$\mathcal{F}_{n,k}(\boldsymbol{\theta}^1_n) \le \mathcal{F}_{n,k}(\boldsymbol{\theta}_n) - \delta \left(1 - \frac{L\delta}{2}\right) \|\nabla \mathcal{F}_{n,k}(\boldsymbol{\theta}_n)\|^2 . \tag{A.5}$$

Then, by invoking Condition A2 in (A.5) and subtracting $\mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})$ from both sides, the following is obtained:

$$\begin{aligned} &\mathcal{F}_{n,k}(\boldsymbol{\theta}^1_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k}) \\ &\le (1 - \delta(2 - L\delta)\mu) \times \left(\mathcal{F}_{n,k}(\boldsymbol{\theta}_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})\right) . \end{aligned} \tag{A.6}$$

After unfolding the single-step update in (A.6) for i iterations, the linear/geometric convergence of $\mathcal{F}_{n,k}(\boldsymbol{\theta}_n)$ can be expressed as

$$\begin{aligned} &\mathcal{F}_{n,k}(\boldsymbol{\theta}^i_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k}) \\ &\le (1 - \delta(2 - L\delta)\mu)^i \times \left(\mathcal{F}_{n,k}(\boldsymbol{\theta}_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})\right) . \end{aligned} \tag{A.7}$$

Furthermore, based on (A.7), (1) is derived as follows. Let $\eta_{n,k}$ denote the normalized relative error between the initial optimization gap and the gap after i iterations in time slot $t \in \mathcal{T}$. Dividing both sides of (A.7) by $\mathcal{F}_{n,k}(\boldsymbol{\theta}_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})$, the following relation holds:

$$\eta_{n,k} \triangleq \frac{\mathcal{F}_{n,k}(\boldsymbol{\theta}^i_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})}{\mathcal{F}_{n,k}(\boldsymbol{\theta}_n) - \mathcal{F}_{n,k}(\boldsymbol{\theta}^\star_{n,k})} \le (1 - \delta(2 - L\delta)\mu)^i . \tag{A.8}$$

Then, for any time slot $t \in \mathcal{T}$, define $r^{\text{pre}}_{n,k}(t) = 1 - Q_{n,k}(t-1)$ and $r^{\text{post}}_{n,k}(t) = 1 - Q_{n,k}(t)$ as the normalized residual QoS gaps before and after the adaptation process, respectively. By taking $\frac{r^{\text{post}}_{n,k}(t)}{r^{\text{pre}}_{n,k}(t)}$ into (A.8), the following holds:

$$\eta_{n,k} = \frac{r^{\text{post}}_{n,k}(t)}{r^{\text{pre}}_{n,k}(t)} \le (1 - \delta(2 - L\delta)\mu)^i . \tag{A.9}$$

Next, by using the notational substitutions $L \equiv \zeta$, $\delta \equiv \chi$, $\mu \equiv \kappa$, $i \equiv i_{n,k}(t)$, and $1 - (2 - \zeta\chi)\chi\kappa \equiv \varrho$, and then multiplying both sides of (A.9) by $r^{\text{pre}}_{n,k}(t)$, the following is obtained:

$$1 - Q_{n,k}(t) \le (1 - Q_{n,k}(t-1))\, \varrho^{i_{n,k}(t)} \tag{A.10}$$

$$\Leftrightarrow Q_{n,k}(t) \ge 1 - (1 - Q_{n,k}(t-1))\, \varrho^{i_{n,k}(t)}. \tag{A.11}$$

Finally, since (A.11) gives a conservative lower bound on the QoS of DT n after $i_{n,k}(t)$ gradient-based refinement steps, this lower envelope is adopted as the nominal QoS recursion in the system model, i.e.,

$$Q_{n,k}(t) = 1 - (1 - Q_{n,k}(t-1))\, \varrho^{i_{n,k}(t)}, \tag{A.12}$$

which proves (1).

Appendix B
Proof of Theorem 1

By squaring both sides of the delay queue, the following is obtained:

$$\begin{aligned} &\left(Q^{\text{dly}}_n(t+1)\right)^2 = \left\{\max\left[Q^{\text{dly}}_n(t) + T^{\text{tol}}_n(t) - T^{\max}, 0\right]\right\}^2 \\ &\le \left(Q^{\text{dly}}_n(t)\right)^2 + \left(T^{\text{tol}}_n(t) - T^{\max}\right)^2 + 2Q^{\text{dly}}_n(t)\left(T^{\text{tol}}_n(t) - T^{\max}\right) . \end{aligned} \tag{A.13}$$

Similarly, by squaring both sides of the cost queue, the following is obtained:

$$\begin{aligned} &\left(Q^{\text{cst}}_n(t+1)\right)^2 = \left\{\max\left[Q^{\text{cst}}_n(t) + C^{\text{tol}}_n(t) - C^{\max}, 0\right]\right\}^2 \\ &\le \left(Q^{\text{cst}}_n(t)\right)^2 + \left(C^{\text{tol}}_n(t) - C^{\max}\right)^2 + 2Q^{\text{cst}}_n(t)\left(C^{\text{tol}}_n(t) - C^{\max}\right) . \end{aligned} \tag{A.14}$$

By subtracting $\left(Q^{\text{dly}}_n(t)\right)^2$ and $\left(Q^{\text{cst}}_n(t)\right)^2$ from both sides, dividing by 2, and summing over all $n \in \mathcal{N}$, the following is obtained:

$$\begin{aligned} &\Delta(\boldsymbol{\Theta}(t)) = \mathbb{E}[L(\boldsymbol{\Theta}(t+1)) - L(\boldsymbol{\Theta}(t)) \mid \boldsymbol{\Theta}(t)] \\ &\le \frac{1}{2} \sum_{n \in \mathcal{N}} \mathbb{E}\Big[\left(T^{\text{tol}}_n(t) - T^{\max}\right)^2 + \left(C^{\text{tol}}_n(t) - C^{\max}\right)^2 \mid \boldsymbol{\Theta}(t)\Big] \\ &\quad + \mathbb{E}\left[\sum_{n \in \mathcal{N}} Q^{\text{dly}}_n(t)\left(T^{\text{tol}}_n(t) - T^{\max}\right) \mid \boldsymbol{\Theta}(t)\right] \\ &\quad + \mathbb{E}\left[\sum_{n \in \mathcal{N}} Q^{\text{cst}}_n(t)\left(C^{\text{tol}}_n(t) - C^{\max}\right) \mid \boldsymbol{\Theta}(t)\right] \\ &\le G + \mathbb{E}\left[\sum_{n \in \mathcal{N}} Q^{\text{dly}}_n(t)\left(T^{\text{tol}}_n(t) - T^{\max}\right) \mid \boldsymbol{\Theta}(t)\right] \\ &\quad + \mathbb{E}\left[\sum_{n \in \mathcal{N}} Q^{\text{cst}}_n(t)\left(C^{\text{tol}}_n(t) - C^{\max}\right) \mid \boldsymbol{\Theta}(t)\right], \end{aligned} \tag{A.15}$$

where the last inequality follows from the finite upper bounds $T^{\text{tol}}_n(t) \le \overline{T}_n$ and $C^{\text{tol}}_n(t) \le \overline{C}_n$.

Lastly, by adding $-V\mathbb{E}[\sum_{n \in \mathcal{N}} \sum_{k \in \mathcal{K}} \omega_{n,k}(t) Q_{n,k}(t) \mid \boldsymbol{\Theta}(t)]$ to both sides of (A.15), inequality (7) can be derived.

APPENDIX C
DETAILED PROOF OF THEOREM 2

In this appendix, a rigorous mathematical derivation of the closed-form forgetfulness model is provided. The proof proceeds in three main steps. First, the excess risk is expressed as the trace of the product of projection matrices. Second, the expectation over Haar-distributed orthogonal matrices is evaluated through Weingarten calculus. Third, an asymptotic expansion of the resulting hypergeometric terms is carried out for large ambient dimension.

A. General Form of Geometric Error

Consider two tasks characterized by their data matrices $\mathbf{X}_A$ and $\mathbf{X}_B$. In the linearized regime, the expected forgetting risk R corresponds to the residual energy of task A projected onto the orthogonal complement of task B's solution space. According to the foundational geometric analysis, for random tasks related by an orthogonal transformation $\mathbf{Q}$, the expected risk can be given by:

$$R(p,m,d) = \mathbb{E}_{\mathbf{Q}\sim\text{Haar}(\mathcal{O}(p))}\left[\text{Tr}(\mathbf{P}_A(\mathbf{I}-\mathbf{P}_B)\mathbf{P}_A)\right], \quad \text{(A.16)}$$

where $\mathbf{P}_A$ and $\mathbf{P}_B$ are projection matrices onto the subspaces of task A and B, respectively. By leveraging the properties of the Weingarten function for integrating polynomials over the orthogonal group, the exact expectation is obtained:

$$R(p,m,d) = \frac{m(p-m)d(p-d)}{(p-1)p(p+2)} \cdot \mathcal{H}(p,m,d), \quad \text{(A.17)}$$

where $\mathcal{H}(p,m,d)$ is a hypergeometric correction factor that accounts for the finite-size effects of the overlap.

B. Asymptotic Expansion and Variable Substitution

Since the exact form involves complex ratios of gamma functions that are analytically opaque for network optimization, an asymptotic expansion for the regime where the network width p is sufficiently large compared to the task dimension d. Substituting the defined physical variables, where $\beta \equiv \beta_{n,j}(t)$, $L^{\min} \equiv L^{\min}_{n,j}$, and $\alpha \equiv \alpha_{k,j}$, i.e., $p \triangleq \gamma\beta$, $d \triangleq \gamma L^{\min}$, and $m \triangleq \gamma\alpha L^{\min}$, into the expansion yields:

$$p \triangleq \gamma\beta, \quad d \triangleq \gamma L^{\min}, \quad m \triangleq \gamma\alpha L^{\min}. \quad \text{(A.18)}$$

Note that the structural density γ elegantly cancels out in the first-order ratios, validating the scale-invariance of our derivation. Defining the normalized load $\rho = L^{\min}/\beta$ and the coupling intensity $\eta = \alpha L^{\min}/\beta$, the term $R(\cdot)$ is expanded in powers of ρ^{-1}. Focusing on the polynomial structure, the resulting terms are grouped by powers of $(1-\rho)$:

- The third-order interaction term $(1-\rho)^3$ arises from the volume correction of the intersection of three subspaces (task A, task B, and the null space). Its coefficient is derived as:
$$C_3 = 5\eta^3 - 18\eta^2\rho^{-1} + 20\eta\rho^{-2} - 6. \quad \text{(A.19)}$$
Substituting $\eta = \alpha\rho$ directly yields the Ψ_3 expression.
- Similarly, the second-order $(1-\rho)^2$ and first-order $(1-\rho)$ terms are systematically extracted from the lower-order cumulants of the weingarten expansion.

Finally, by normalizing the risk by the total parameter volume β and discarding negligible higher-order infinitesimals ($\mathcal{O}(1/p^2)$), the closed-form expression in (8) is obtained. This expression retains the essential non-linear dependencies on α and β, providing a precise yet tractable objective for the optimization problem.

APPENDIX D
PROOF OF THEOREM 3

For iteration r, the constructed surrogate satisfies

$$\mathcal{V}_t(\mathbf{x}) \le \hat{\mathcal{V}}_t(\mathbf{x};\mathbf{x}^{(r)}), \quad \mathcal{V}_t(\mathbf{x}^{(r)}) = \hat{\mathcal{V}}_t(\mathbf{x}^{(r)};\mathbf{x}^{(r)}). \quad \text{(A.20)}$$

Since $\mathbf{x}^{(r+1)} = \arg\min_{\mathbf{x}} \hat{\mathcal{V}}_t(\mathbf{x};\mathbf{x}^{(r)})$, it holds that

$$\begin{aligned}\mathcal{V}_t(\mathbf{x}^{(r+1)}) &\le \hat{\mathcal{V}}_t(\mathbf{x}^{(r+1)};\mathbf{x}^{(r)}) \\ &\le \hat{\mathcal{V}}_t(\mathbf{x}^{(r)};\mathbf{x}^{(r)}) = \mathcal{V}_t(\mathbf{x}^{(r)}). \quad \text{(A.21)}\end{aligned}$$

Hence, $\{\mathcal{V}_t(\mathbf{x}^{(r)})\}$ is monotonically non-increasing. Because the relaxed feasible set of $\mathcal{P}_2$ is compact and $\mathcal{V}_t(\mathbf{x})$ is lower bounded, the sequence converges. In addition, the surrogate and the relaxed objective are first-order consistent at every iterate, so every accumulation point of $\{\mathbf{x}^{(r)}\}$ satisfies the KKT condition of the relaxed per-slot problem.

APPENDIX E
PROOF OF THEOREM 5

For each DT n, the surrogate problem contains K relaxed update variables $\tilde{u}_{n,k}(t)$, K task-specific layer variables $\tilde{\beta}_{n,k}(t)$, K auxiliary variables $y_{n,k}(t)$, and two resource variables $b_n(t)$ and $f_n(t)$. Hence, the number of decision variables is

$$n_{\text{var}} = N(3K+2) = \mathcal{O}(NK). \quad \text{(A.22)}$$

The constraints consist of N update-selection inequalities, $2NK$ box constraints on $\tilde{u}_{n,k}(t)$, $2NK$ box constraints on $\tilde{\beta}_{n,k}(t)$, $4NK$ McCormick inequalities, and 4 global resource and trust-region constraints. Therefore,

$$m_{\text{constr}} = N + 8NK + 4 = \mathcal{O}(NK). \quad \text{(A.23)}$$

By the standard complexity bound of primal-dual interior-point methods, the cost of each SCA iteration is

$$\mathcal{O}\left(\sqrt{m_{\text{constr}}}\, n_{\text{var}}^3\right) = \mathcal{O}((NK)^{3.5}). \quad \text{(A.24)}$$

Since the rounding and feasibility-restoration steps require at most $\mathcal{O}(NK)$ operations, the dominant per-slot cost of SOUND is $\mathcal{O}\left(I_{\text{SCA}} \cdot (NK)^{3.5}\right)$.

APPENDIX F
PROOF OF THEOREM 4

The finite approximation error used in Theorem 4 is first established. Let $\hat{\mathcal{V}}_t^{\text{sca}}$ and $\hat{\mathcal{V}}_t^{\text{rnd}}$ denote the objective values after the SCA stage and the randomized rounding stage, respectively. Then, the total per-slot error can be decomposed as

$$\hat{\mathcal{V}}_t^{\text{final}} - \mathcal{V}_t^* = \left(\hat{\mathcal{V}}_t^{\text{sca}} - \mathcal{V}_t^*\right) + \left(\hat{\mathcal{V}}_t^{\text{rnd}} - \hat{\mathcal{V}}_t^{\text{sca}}\right) + \left(\hat{\mathcal{V}}_t^{\text{final}} - \hat{\mathcal{V}}_t^{\text{rnd}}\right). \quad \text{(A.1)}$$

Taking the conditional expectation yields

$$\begin{aligned}&\mathbb{E}[\hat{\mathcal{V}}_t^{\text{final}} - \mathcal{V}_t^* \mid \mathbf{\Theta}(t)] \\ &\le \mathbb{E}[\Delta_t^{\text{sca}} \mid \mathbf{\Theta}(t)] + \mathbb{E}[\Delta_t^{\text{rnd}} \mid \mathbf{\Theta}(t)] + \mathbb{E}[\Delta_t^{\text{res}} \mid \mathbf{\Theta}(t)].\end{aligned} \quad \text{(A.2)}$$

The relaxed feasible set is compact because every task-specific layer decision $\tilde{\beta}_{n,k}(t)$ lies in $[L^{\min}_{n,k}, L^{\max}_{n,k}]$, while the update and resource variables are also bounded. Hence, there exist finite constants Δ^{sca}, Δ^{rnd}, and Δ^{res} such that

$$\begin{aligned}\mathbb{E}[\Delta^{\text{sca}}_t \mid \mathbf{\Theta}(t)] &\le \Delta^{\text{sca}},\\ \mathbb{E}[\Delta^{\text{rnd}}_t \mid \mathbf{\Theta}(t)] &\le \Delta^{\text{rnd}},\\ \mathbb{E}[\Delta^{\text{res}}_t \mid \mathbf{\Theta}(t)] &\le \Delta^{\text{res}}.\end{aligned} \tag{A.3}$$

By defining $\Delta_{\text{approx}} \triangleq \Delta^{\text{sca}} + \Delta^{\text{rnd}} + \Delta^{\text{res}}$, the following is obtained:

$$\mathbb{E}[\hat{\mathcal{V}}^{\text{final}}_t - \mathcal{V}^*_t \mid \mathbf{\Theta}(t)] \le \Delta_{\text{approx}}. \tag{A.4}$$

Let $\mathbf{x}^*(t)$ be an exact minimizer of the $\mathcal{P}_2$, and let $\mathbf{x}^{\text{SOUND}}(t)$ be the feasible per-slot decision generated by SOUND at time slot t. Since the terms $-\sum_{n\in\mathcal{N}} Q^{\text{dly}}_n(t)T^{\max}$ and $-\sum_{n\in\mathcal{N}} Q^{\text{cst}}_n(t)C^{\max}$ in (7) are independent of the decision variables, minimizing the right-hand side of (7) is equivalent to minimizing the objective $\mathcal{V}_t(\mathbf{x})$ of $\mathcal{P}_2$. Since $\hat{\mathcal{V}}^{\text{final}}_t = \mathcal{V}_t(\mathbf{x}^{\text{SOUND}}(t))$, the following holds:

$$\mathbb{E}[\mathcal{V}_t(\mathbf{x}^{\text{SOUND}}(t)) \mid \mathbf{\Theta}(t)] \le \mathbb{E}[\mathcal{V}_t(\mathbf{x}^*(t)) \mid \mathbf{\Theta}(t)] + \Delta_{\text{approx}}. \tag{A.5}$$

From stochastic network optimization theory, if problem $\mathcal{P}_1$ is stabilizable, then there exists a stationary randomized policy π^* that depends only on the exogenous system state in each slot, such as $\omega_{n,k}(t)$, $h_{n,m}(t)$, $D^p_{n,k}(t)$, $\lambda_n(t)$, and $\alpha_{k,j}$, but not on the virtual queue backlogs. This policy achieves the optimal priority-weighted system QoS $\mathcal{Q}^{\text{opt}}$ and satisfies

$$\mathbb{E}[T^{\text{tol}}_n(\pi^*)] \le T^{\max}, \quad \mathbb{E}[C^{\text{tol}}_n(\pi^*)] \le C^{\max}, \quad \forall n. \tag{A.6}$$

Since $\mathbf{x}^*(t)$ minimizes $\mathcal{V}_t(\mathbf{x})$ over the feasible set of $\mathcal{P}_2$, it follows that

$$\mathbb{E}[\mathcal{V}_t(\mathbf{x}^*(t)) \mid \mathbf{\Theta}(t)] \le \mathbb{E}[\mathcal{V}_t(\pi^*) \mid \mathbf{\Theta}(t)]. \tag{A.7}$$

Moreover, the per-slot objective under π^* can be written as

$$\begin{aligned}\mathbb{E}[\mathcal{V}_t(\pi^*) \mid \mathbf{\Theta}(t)] &= \sum_{n\in\mathcal{N}} Q^{\text{dly}}_n(t)\mathbb{E}[T^{\text{tol}}_n(\pi^*) \mid \mathbf{\Theta}(t)]\\ &+\sum_{n\in\mathcal{N}} Q^{\text{cst}}_n(t)\mathbb{E}[C^{\text{tol}}_n(\pi^*) \mid \mathbf{\Theta}(t)] - V\mathbb{E}[\mathcal{Q}^{\text{sys}}(\pi^*) \mid \mathbf{\Theta}(t)].\end{aligned} \tag{A.8}$$

Substituting $\mathbf{x}^{\text{SOUND}}(t)$ into (7), and then applying the above two inequalities, the following is obtained:

$$\begin{aligned}&\Delta_V(\mathbf{\Theta}(t))\\ &\le G - \sum_{n\in\mathcal{N}} Q^{\text{dly}}_n(t)T^{\max} - \sum_{n\in\mathcal{N}} Q^{\text{cst}}_n(t)C^{\max} + \\ &\quad \mathbb{E}[\mathcal{V}_t(\mathbf{x}^{\text{SOUND}}(t)) \mid \mathbf{\Theta}(t)]\\ &\le G + \Delta_{\text{approx}} - \sum_{n\in\mathcal{N}} Q^{\text{dly}}_n(t)T^{\max}\\ &\quad - \sum_{n\in\mathcal{N}} Q^{\text{cst}}_n(t)C^{\max} + \mathbb{E}[\mathcal{V}_t(\pi^*) \mid \mathbf{\Theta}(t)]\\ &= G + \Delta_{\text{approx}} + \sum_{n\in\mathcal{N}} Q^{\text{dly}}_n(t)\mathbb{E}[T^{\text{tol}}_n(\pi^*) - T^{\max} \mid \mathbf{\Theta}(t)]\\ &\quad + \sum_{n\in\mathcal{N}} Q^{\text{cst}}_n(t)\mathbb{E}[C^{\text{tol}}_n(\pi^*) - C^{\max} \mid \mathbf{\Theta}(t)]\\ &\quad - V\mathbb{E}[\mathcal{Q}^{\text{sys}}(\pi^*) \mid \mathbf{\Theta}(t)]\\ &\le G + \Delta_{\text{approx}} - V\mathcal{Q}^{\text{opt}}.\end{aligned} \tag{A.9}$$

The last inequality follows because the queue-weighted delay and cost terms are non-positive under π^*.

Taking the total expectation of both sides, summing over $t = 0, \ldots, T-1$, and using the telescoping property of the Lyapunov drift yields

$$\begin{aligned}&\mathbb{E}[L(\mathbf{\Theta}(T))] - \mathbb{E}[L(\mathbf{\Theta}(0))] - V\sum_{t=0}^{T-1}\mathbb{E}[\mathcal{Q}^{\text{sys}}(t)]\\ &\le T(G + \Delta_{\text{approx}} - V\mathcal{Q}^{\text{opt}}).\end{aligned} \tag{A.10}$$

Rearranging and dividing both sides by VT give

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\mathcal{Q}^{\text{sys}}(t)] \ge \mathcal{Q}^{\text{opt}} - \frac{G+\Delta_{\text{approx}}}{V} - \frac{\mathbb{E}[L(\mathbf{\Theta}(T)) - L(\mathbf{\Theta}(0))]}{VT}. \tag{A.11}$$

Since $L(\mathbf{\Theta}(T)) \ge 0$ and the initial Lyapunov value is finite, the last term vanishes as $T \to \infty$, which proves the stated optimality-gap bound.

To derive the queue-backlog bound, let π^ϵ be a stationary randomized policy satisfying Slater's condition with some $\epsilon > 0$, i.e.,

$$\mathbb{E}[T^{\text{tol}}_n(\pi^\epsilon)] \le T^{\max} - \epsilon, \mathbb{E}[C^{\text{tol}}_n(\pi^\epsilon)] \le C^{\max} - \epsilon, \forall n. \tag{A.12}$$

Since $Q_{n,k}(t) \in [0,1]$ and the priorities are finite in every time slot, there exists a finite constant $\overline{\mathcal{Q}}$ such that $\mathcal{Q}^{\text{sys}}(t) \le \overline{\mathcal{Q}}$ for all t. Repeating the same argument with π^ϵ gives

$$\begin{aligned}\Delta_V(\mathbf{\Theta}(t)) \le& G + \Delta_{\text{approx}} - \epsilon\sum_{n\in\mathcal{N}}\left(Q^{\text{dly}}_n(t) + Q^{\text{cst}}_n(t)\right)\\ &- V\mathbb{E}[\mathcal{Q}^{\text{sys}}(\pi^\epsilon) \mid \mathbf{\Theta}(t)].\end{aligned} \tag{A.13}$$

Using $\Delta_V(\mathbf{\Theta}(t)) = \Delta(\mathbf{\Theta}(t)) - V\mathbb{E}[\mathcal{Q}^{\text{sys}}(t) \mid \mathbf{\Theta}(t)]$, $\mathcal{Q}^{\text{sys}}(t) \le \overline{\mathcal{Q}}$, and $\mathcal{Q}^{\text{sys}}(\pi^\epsilon) \ge 0$, the following is further obtained:

$$\Delta(\mathbf{\Theta}(t)) \le G + \Delta_{\text{approx}} + V\overline{\mathcal{Q}} - \epsilon\sum_{n\in\mathcal{N}}\left(Q^{\text{dly}}_n(t) + Q^{\text{cst}}_n(t)\right). \tag{A.14}$$

For brevity, define $B_\Theta(t) \triangleq \sum_{n\in\mathcal{N}}\mathbb{E}[Q^{\text{dly}}_n(t) + Q^{\text{cst}}_n(t)]$. Summing the above inequality over $t = 0, \ldots, T-1$ yields

$$\begin{aligned}\epsilon\sum_{t=0}^{T-1} B_\Theta(t) \le& T(G + \Delta_{\text{approx}} + V\overline{\mathcal{Q}})\\ &+ \mathbb{E}[L(\mathbf{\Theta}(0))] - \mathbb{E}[L(\mathbf{\Theta}(T))].\end{aligned} \tag{A.15}$$

Dividing both sides by ϵT and taking $\limsup_{T\to\infty}$ produces

$$\limsup_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1}\sum_{n\in\mathcal{N}}\mathbb{E}[Q^{\text{dly}}_n(t) + Q^{\text{cst}}_n(t)] \le \frac{G + \Delta_{\text{approx}} + V\overline{\mathcal{Q}}}{\epsilon}, \tag{A.16}$$

which completes the proof.